USN | | | | | | | | | | [22ME53]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

HEAT AND MASS TRANSFER

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. HMT data book permitted for the examination

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	Write down the general equation for one dimensional steady state heat transfer in slab or plane wall with and without heat generation	L2	CO1	**02**
	b.	Discuss the significance of the Grashof number.	L2	CO2	**02**
	c.	Define the properties emissivity and absorptivity. When are these two properties equal to each other?	L2	CO3	**02**
	d.	What is the difference between evaporation and boiling?	L2	CO4	**02**
	e.	State Fick's law of diffusion.	L1	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Discuss on the lumped parameter analysis.	L2	CO1	**05**
		OR			
	b.	With relevant examples, explain the mechanisms of conduction, convection and radiation heat transfer.	L2	CO1	**05**
3.	**a.**	How the formation of free convection boundary layer that develops on a heated vertical plate.	L3	CO2	**05**
		OR			
	b.	Air at 20°C, at a pressure of 1 bar is flowing over a flat plate at a velocity of 3 m/s. if the plate maintained at 60°C, calculate the heat transfer per unit width of the plate. Assuming the length of the plate along the flow of air is 2 m.	L3	CO2	**05**
4.	**a.**	Explain the concept of absorptivity, reflectivity and transmissivity.	L2	CO3	**05**
		OR			
	b.	Explain the concept of black body and gray body.	L2	CO3	**05**
5.	**a.**	When the outlet temperatures of the fluids in a heat exchanger are not known, is it still practical to use the LMTD method? Explain.	L3	CO4	**05**
		OR			

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	b.	Calculate the power required to boil water in a copper pan, 0.35 m in diameter. The pan is maintained at 120 °C by an electric heater.	L3	CO4	**05**
6.	**a.**	Demonstrate the analogy between heat and mass transfer.	L3	CO5	**05**
		OR			
	b.	Gaseous hydrogen diffuses through a steel wall of 3 mm thickness. The molar concentration of hydrogen at the interface is 1.5 kg mol/m^3 and it is zero on the outer face. Determine the diffusion rate of hydrogen.	L3	CO5	**05**
7.	**a.**	What is critical radius/thickness of insulation? Explain its significance for insulation of cylinder and sphere.	L2	CO1	**05**
		OR			
	b.	Explain the concept of thermal and hydrodynamic boundary layer.	L2	CO2	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Discuss the general governing differential equation and develop heat dissipation equation for a short fin with its end insulated.	L3	CO1	**10**
		OR			
	b.	A furnace wall made up of 7.5 cm of fire plate and 0.65 cm of mild steel plate. Inside surface exposed to hot gas at 650°C and outside air temperature 27°C. The convective heat transfer co-efficient for inner side is 60 W/m^{2}K. The convective heat transfer co-efficient for outer side is 8 W/m^{2}K. Calculate the heat lost per square meter area of the furnace wall and also find outside surface temperature.	L3	CO1	**10**
9.	**a.**	Air at 20 °C at atmospheric pressure flows over a flat plate at a velocity of 3 m/s. if the plate is 1 m wide and 80°C, calculate the following at x = 300 mm. i. Hydrodynamic boundary layer thickness, ii. Thermal boundary layer thickness, iii. Local friction coefficient, iv. Average friction coefficient, v. Local heat transfer coefficient vi. Average heat transfer coefficient, vii. Heat transfer.	L3	CO2	**10**
		OR			
	b.	Engine oil flows through a 50 mm diameter tube at an average temperature of 147 °C. The flow velocity is 80 cm/s. Calculate	L3	CO2	**10**

		the average heat transfer coefficient if the tube wall is maintained at a temperature of 200 °C and it is 2 m long.			
10.	**a.**	Determine net radiation heat transfer per m^2 for two infinite parallel plates held at temperature of 800 K and 500 K respectively. Emissivities of hot and cold plates are 0.6 and 0.4 respectively. Now it is intended to reduce the heat transfer to 40% of original value by placing a radiation shield between the plates. Calculate the emissivity of the shield and its equilibrium temperature.	L3	CO3	**10**
		OR			
	b.	Calculate the following for an industrial furnace in the form of a black body and emitting radiation of 2500 °C : **i.** Monochromatic emissive power at 1.2 μm length, **ii.** Wavelength at which the emission is maximum, **iii.** Maximum emissive power, **iv.** Total emissive power, and **v.** Total emissive power of the furnace if it is assumed as a real surface with emissivity equal to 0.9.	L3	CO2	**10**
11.	**a.**	Derive expression to find logarithmic mean temperature difference for parallel flow heat exchanger.	L3	CO4	**10**
		OR			
	b.	In a counter flow double pipe heat exchanger, water is heated from 25 °C to 65 °C by oil with specific heat of 1.45 kJ/kg K and mass flow rate of 0.9 kg/s. The oil is cooled from 230 °C to 160 °C. If overall Heat transfer coefficient is 420 W/m^2 °C. calculate following: a) The rate of heat transfer b) The mass flow rate of water, and c) The surface area of heat exchanger	L3	CO3	**10**
12.	**a.**	Hydrogen gases at 3 bar and 1 bar are separated by a plastic membrane having thickness 0.25 mm. the binary diffusion coefficient of hydrogen in the plastic is 9.1×10^{-3} m^2/s. The solubility of hydrogen in the membrane is 2.1×10^{-3} kg-mole/m^3.bar. An uniform temperature condition of 20 °C is assumed. Calculate the following 1. Molar concentration of hydrogen on both sides 2. Molar flux of hydrogen 3. Mass flux of hydrogen	L3	CO5	**10**
		OR			
	b.	A vessel contains a binary mixture of O_2 and N_2 with partial	L3	CO5	**10**

		pressures in the ratio 0.21 and 0.79 at 20 °C. If the total pressure of the mixture is 1.1 bar, calculate the following: **i.** Molar concentrations **ii.** Mass densities **iii.** Mass fractions **iv.** Molar fractions of each species			
13.	**a.**	Two long rods of the same diameter, one made of brass (k=85 W/m°C) and the other of copper (k= 375 W/m°C), have one of their ends inserted into a furnace. Both of the rods are exposed to the same environment. At a distance 105 mm away from the furnace end, the temperature of the brass rod is 120°C. at what distance from the furnace end, the same temperature would be reached in the copper rod?	L4	CO6	**10**
		OR			
	b.	In an open-heart surgery under hypothermic conditions, the patient's blood is cooled before the surgery and rewarded afterwards. It is proposed that a concentric tube flow heat exchanger of length 0.5 m is to be used for this purpose with a thin-walled inner tube having a diameter of 55 mm. If water at 60°C and 0.1 kg/s is used to heat blood entering the heat exchanger at 18°C and 0.05kg/s, what is the temperature of blood leaving the heat exchanger and the heat flow rate. Take Uo = 500 W/m^2 K,Cp of blood = 3.5 kJ/kg K.	L4	CO6	**10**